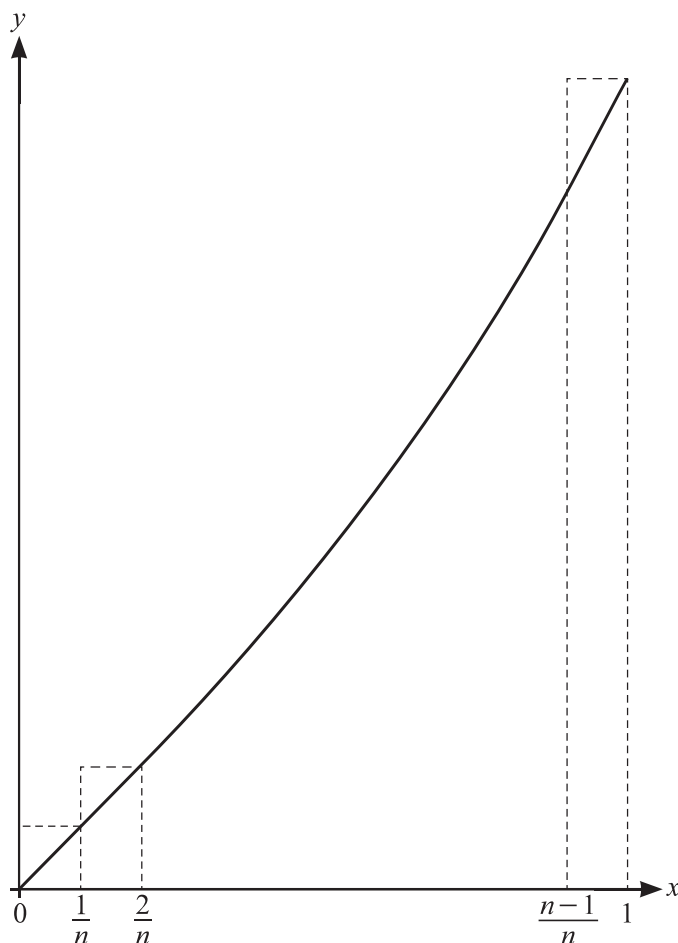




The diagram shows the curve with equation $y = \frac{1}{3}x^3 + x$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

(a) By considering the sum of the areas of these rectangles, show that $\int_0^1 \left(\frac{1}{3}x^3 + x \right) dx < U_n$, where

$$U_n = \frac{1}{12} \left(1 + \frac{1}{n} \right) \left(7 + \frac{1}{n} \right). \quad [5]$$

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- (b) Use a similar method to find, in terms of n , a lower bound L_n for $\int_0^1 \left(\frac{1}{3}x^3 + x\right) dx$. [4]

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- (c) Show that $\lim_{n \rightarrow \infty} (U_n - L_n) = 0$. [2]

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